A Comparative Survey of Attention Mechanisms in Deep Neural Networks

Abstract

This paper surveys attention mechanisms used in modern deep learning architectures, comparing additive, multiplicative, and self-attention variants. We evaluate their computational cost, representational power, and empirical performance across NLP and vision tasks, and propose a taxonomy that clarifies design trade-offs for practitioners.

1. Introduction

Attention mechanisms have become a cornerstone of modern neural network design, enabling models to selectively focus on relevant parts of their input. Since the introduction of the Transformer architecture, self-attention has largely superseded recurrent approaches for sequence modeling. This survey systematically reviews the evolution of attention from early additive formulations to scaled dot-product self-attention, and discusses hybrid approaches that combine attention with convolutional or recurrent inductive biases.

2. Methodology

We conduct a systematic literature review of 45 papers published between 2015 and 2024, extracting architectural details, benchmark datasets, and reported metrics. Each attention variant is re-implemented in a shared PyTorch framework and benchmarked on WMT14 English-German translation, ImageNet-1k classification, and GLUE for controlled comparison. Training used identical hyperparameter budgets across all variants to isolate architectural effects.

3. Results

Scaled dot-product self-attention achieved the best BLEU score of 28.4 on WMT14, outperforming additive attention (26.1 BLEU) while requiring 35 percent less training time due to parallelization. On ImageNet-1k, vision transformer variants using windowed self-attention matched convolutional baselines at 81.2 percent top-1 accuracy with fewer FLOPs at inference time.

4. Limitations

Our benchmarks were restricted to English-centric datasets, and the computational comparisons assumed access to modern GPU hardware; results may not generalize to edge deployment scenarios or low-resource languages. Additionally, we did not evaluate emerging linear-attention approximations at scale beyond 1 billion parameters.

5. Future Work

Future research should explore hardware-aware attention variants for edge devices, extend benchmarking to multilingual and low-resource settings, and investigate the interpretability of learned attention weights in safety-critical applications such as medical diagnosis.

References

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